Laws of Motion - Practice Questions

1. A 50 N retarding force acts on a 20 kg body moving at 15 m/s. Time to stop?

Given: F = 50 N (retarding), m = 20 kg, u = 15 m/s, v = 0

Retardation $a = F / m = \frac{50}{20} = 2.5\ m/s^2$

Using $v = u - at$ with v = 0:

$t = u / a = \frac{15}{2.5} = 6\ s$

2. A 5 kg block rests on a rough surface (μ = 0.4, g = 10). Minimum force to start it moving?

Given: m = 5 kg, μ = 0.4, g = 10 m/s²
Normal reaction $N = mg = 5 \times 10 = 50\ N$
Limiting friction $f = \mu N = 0.4 \times 50 = 20\ N$
Minimum force to just move the block $F = f = 20\ N$

3. A batsman deflects a 0.15 kg ball by 45° without changing its speed of 54 km/h. Impulse?

Given: m = 0.15 kg, speed = 54 km/h, deflection = 45°

$v = 54 \times \frac{5}{18} = 15\ m/s$

The incoming and outgoing paths each make 22.5° with the line of impulse.
Momentum components along that line add; perpendicular ones cancel.
Impulse $J = \Delta p = mv \cos 22.5° + mv \cos 22.5° = 2mv \cos 22.5°$
$J = 2 \times 0.15 \times 15 \times 0.924 \approx 4.16\ N{\cdot}s$

4. Masses 8 kg and 12 kg hang over a frictionless pulley (g = 10). Acceleration and tension?

Given: m1 = 8 kg, m2 = 12 kg, g = 10 m/s². Heavier mass (12 kg) moves down.
For 12 kg: $m2\,g - T = m2\,a \rightarrow 120 - T = 12a$ ...(1)
For 8 kg: $T - m1\,g = m1\,a \rightarrow T - 80 = 8a$ ...(2)
Adding (1) and (2): $40 = 20a$

$a = \frac{40}{20} = 2\ m/s^2$

Put a = 2 in (2): $T = 80 + 8 \times 2 = 96\ N$

Check with formulas:

$a = \frac{(m2 - m1)\,g}{m1 + m2} = \frac{(12 - 8) \times 10}{20} = 2\ m/s^2$

$$T = \frac{2\,m1\,m2\,g}{m1 + m2} = \frac{2 \times 8 \times 12 \times 10}{20} = 96\ N$$

5. A 0.25 kg stone is whirled in a horizontal circle of radius 1.5 m at 40 rev/min. Tension? Max speed if the string can take 200 N?

Given: m = 0.25 kg, r = 1.5 m, n = 40 rev/min, T_max = 200 N

$$n = 40\ \text{rev/min} = \frac{40}{60}\ \text{rev/s} = 2/3\ \text{rev/s}$$

$$\omega = 2\pi n = 2\pi \times \frac{2}{3} = \frac{4\pi}{3}\ \text{rad/s} \approx 4.19\ \text{rad/s}$$

Tension provides the centripetal force:

$T = m\omega^2 r = 0.25 \times (4.19)^2 \times 1.5 = 0.25 \times 17.55 \times 1.5 \approx 6.6\ N$

For max speed, set T = 200 N in $T = mv^2 / r$:

$$v_{max}^2 = \frac{T\,r}{m} = \frac{200 \times 1.5}{0.25} = 1200$$

$v_{max} = \sqrt{1200} \approx 34.6\ m/s$

6. A 1000 kg helicopter rises at 15 m/s² carrying 300 kg of crew (g = 10). Find:

Given: M = 1000 kg, m = 300 kg, a = 15 m/s² upward, g = 10 m/s²

(a) Force on floor by crew

Floor pushes crew up with N; $N - mg = ma$

$N = m(g + a) = 300 \times (10 + 15) = 300 \times 25 = 7500\ N$

By Newton's 3rd law, crew push floor with 7500 N, downward

(b) Action of rotor on air

Air lifts the whole system (helicopter + crew) with force F; $F - (M + m)g = (M + m)a$

$F = (M + m)(g + a) = 1300 \times 25 = 32\,500\ N$

By Newton's 3rd law, rotor pushes air with 32 500 N, downward

(c) Force on helicopter by air

Reaction to (b): 32 500 N, upward